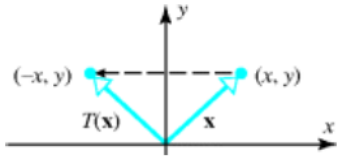
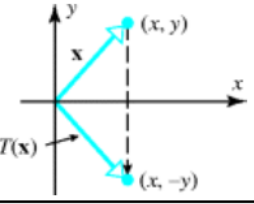
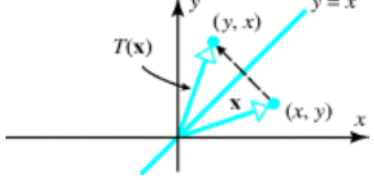


2- Reflection

Let us find the images of the standard basis vectors e_1, e_2 for R^2 in column form.

Table 1

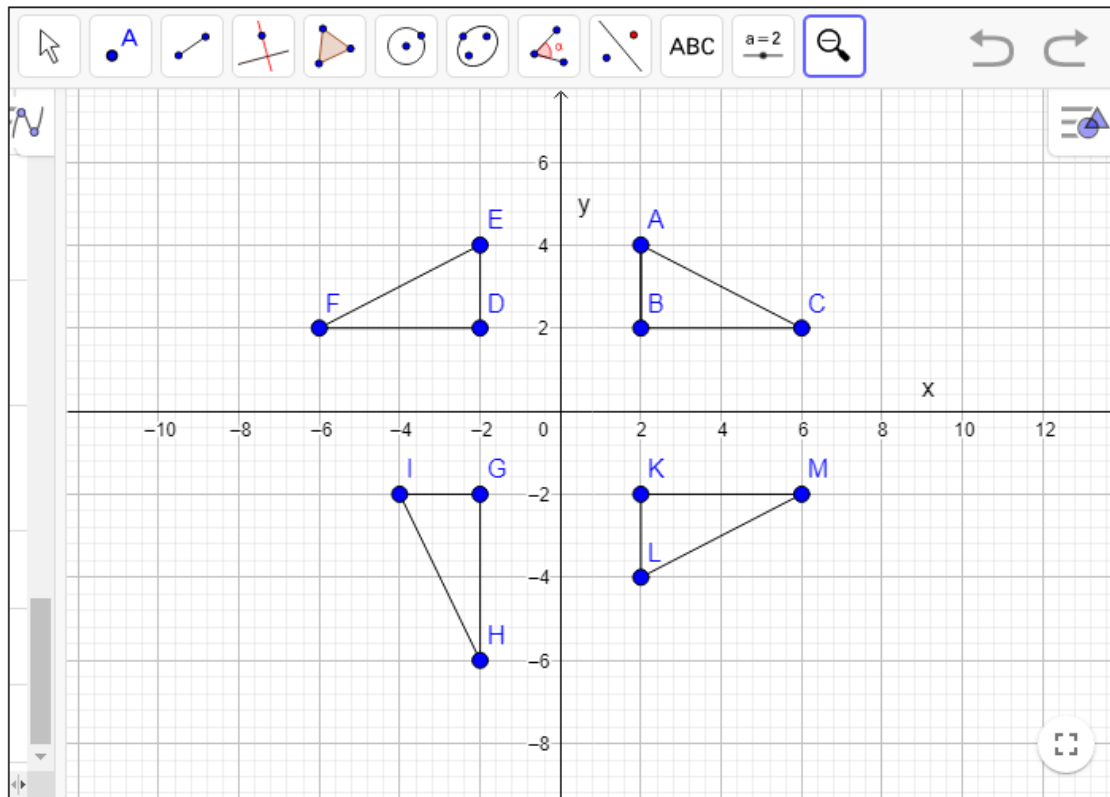
Operator	Illustration	Images of e_1 and e_2	Standard Matrix
Reflection about the y -axis $T(x, y) = (-x, y)$		$T(e_1) = T(1, 0) = (-1, 0)$ $T(e_2) = T(0, 1) = (0, 1)$	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection about the x -axis $T(x, y) = (x, -y)$		$T(e_1) = T(1, 0) = (1, 0)$ $T(e_2) = T(0, 1) = (0, -1)$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection about the line $y = x$ $T(x, y) = (y, x)$		$T(e_1) = T(1, 0) = (0, 1)$ $T(e_2) = T(0, 1) = (1, 0)$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Activate M

Reflection about the line $y = -x$ $T(x, y) = (-y, -x)$	$T(e_1) = T(1, 0) = (0, -1)$ $T(e_2) = T(0, 1) = (-1, 0)$	$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
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Example 1. (Reflection of Triangle)

Reflect the triangle with vertices $A = (2, 4)$, $B = (2, 2)$, $C = (6, 2)$ along x-axis, y-axis and $y = -x$.



Example 2. (Reflection of a line)

Let $y = 2x + 1$ be a line. Find the reflection of that line along the line $y = x$.

Solution. Here

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Therefore, $T(\vec{x}) = A\vec{x}$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$$

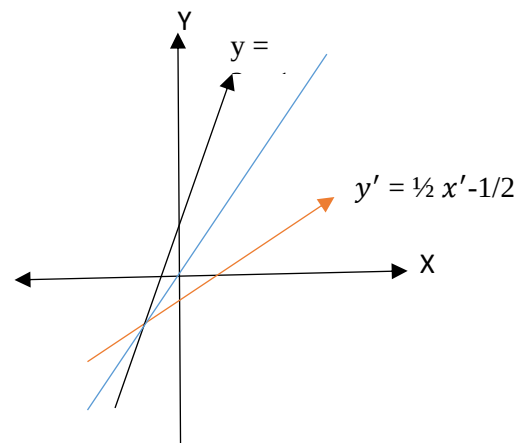
So, $x = y'$, $y = x'$

Put x and y in original line $y = 2x + 1$

$$x' = 2y' + 1$$

$$\text{Or } 2y' = x' - 1$$

So $y' = \frac{1}{2}x' - \frac{1}{2}$ is the reflected line.



To draw original line $y = 2x + 1$ take two points on it, let $A = (1, 3)$ and $B = (2, 5)$.

And to draw the Reflected line $y' = \frac{1}{2} x' - \frac{1}{2}$, $A' = (2, 1/2)$ and $B' = (4, 3/2)$.

Reflection of circle

Let $(x-2)^2 + (y-3)^2 = 4$ be a circle. Find its reflection along the line $y = -x$

Solution. The transformation of reflection is

$$T(\vec{x}) = A\vec{x}, \quad \forall \vec{x} \in \mathbb{R}^2$$

Where

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

So,
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ -x \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x' &= -y \\ y' &= -x \end{aligned}$$

Putting $x = -y'$, $y = -x'$ in the original circle $(x - 2)^2 + (y - 3)^2 = 4$, we get

$$(x' + 3)^2 + (y' + 2)^2 = 4, \text{ reflected circle.}$$

As original circle $(x - 2)^2 + (y - 3)^2 = 4$ is with Centre = $(2, 3)$ and Radius = 2

While Reflected circle $(x' + 3)^2 + (y' + 2)^2 = 4$ has Centre = $(-3, -2)$, Radius = 2.

We can draw both circles easily.

3-Rotation

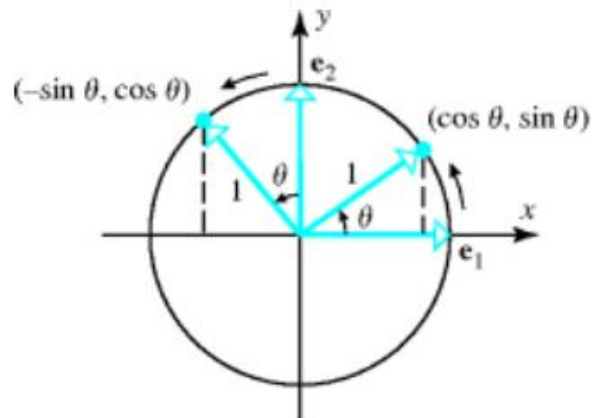
Rotation about origin through an angle θ is a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as:

$$T(\vec{x}) = A\vec{x} : \forall \vec{x} \in \mathbb{R}^2$$

$$T(\mathbf{e}_1) = T(1, 0) = (\cos \theta, \sin \theta) \text{ and } T(\mathbf{e}_2) = T(0, 1) = (-\sin \theta, \cos \theta)$$

Where

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



- If the direction of θ is not defined, then it is understood to be in anticlockwise direction.

- If θ is in clockwise direction, then replace θ by $-\theta$ in the above definition as:

$$A = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

Example 1: Sketch the image of given rectangle with vertices A(0,0), B(3,0), C(3,2), D(0,2) under the rotation of 30° (anticlockwise).

Solution: As the transformation of rotation is

$$T(\vec{x}) = A\vec{x} \quad : \forall \vec{x} \in R^2$$

Where

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

As $\theta = 30^\circ$, so

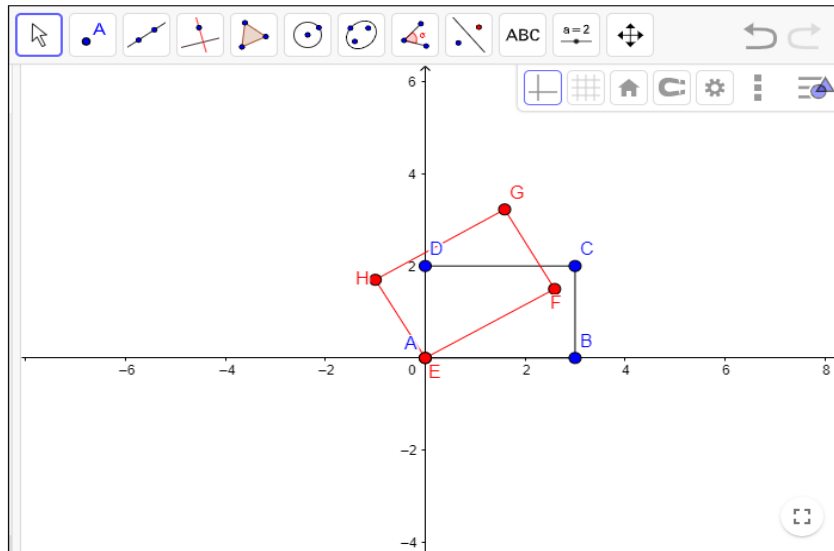
$$A = \begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

For point A: $T \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

For point B: $T \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{3}}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 2.598 \\ 1.5 \end{bmatrix}$

For point C: $T \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{3}}{2} - 1 \\ \frac{3}{2} + \sqrt{3} \end{bmatrix} = \begin{bmatrix} 1.599 \\ 3.23 \end{bmatrix}$

For point D: $T \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1.732 \end{bmatrix}$



Work to do:

Q1. Sketch the image of given parallelogram with vertices A(0,1), B(3,0), C(5,-2), D(2,-1) under the rotation of 90° (anticlockwise) .

Q2. Sketch the image of given triangle with vertices A(2,4), B(2,2), C(4,2) under the rotation of 90° (clockwise) .

Example 2. Let $y = 2x+5$ be a line. Find the equation of line after rotating it through an angle of $\frac{\pi}{2}$ clockwise direction about origin.

Solution: The matrix of rotation in clockwise direction is

$$A = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

As $\theta = \frac{\pi}{2}$, so

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

and

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$$

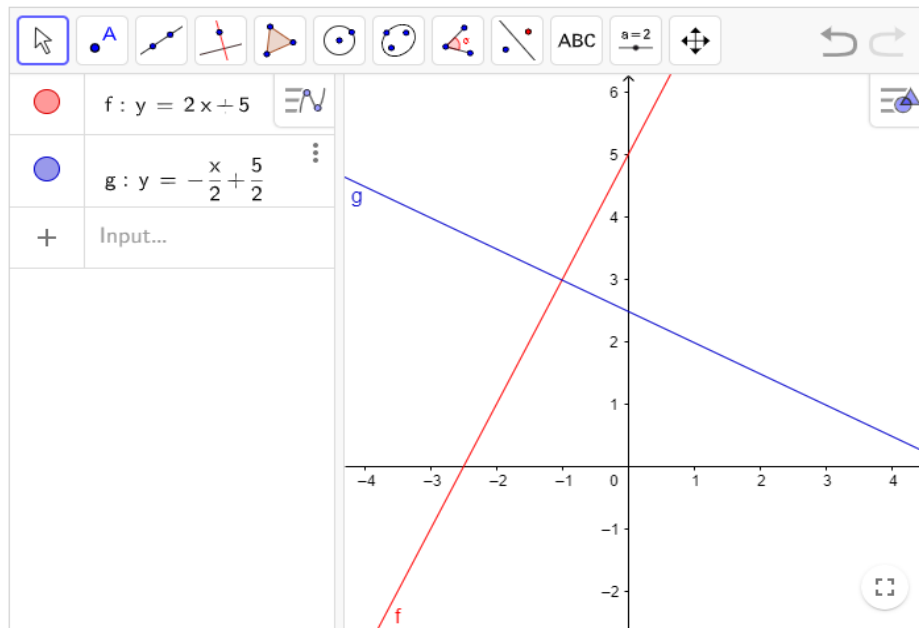
or

$$\begin{cases} x' = y \\ y' = -x \end{cases} \Rightarrow \begin{cases} x = -y' \\ y = x' \end{cases}$$

Put value of x and y in original equation of line $y = 2x + 5$ and obtain

$$y' = -\frac{x'}{2} + \frac{5}{2}$$

This is the rotated line with angle $\frac{\pi}{2}$ in clockwise direction.



Work to do:

Q3. Let $y = -2x + 7$ be a line. Find the equation of line after rotating it through an angle of $180^\circ, 270^\circ$ clockwise direction about origin.

Example 3. Let $(x - 4)^2 + (y - 3)^2 = 9$ be a circle. Find the equation of circle after rotating it through an angle of 90° in anticlockwise direction about origin.

Solution: The matrix of rotation in anticlockwise direction is

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

As $\theta = 90^\circ$, so

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Therefore, for

$$T(\vec{x}) = A\vec{x} : \forall \vec{x} \in \mathbb{R}^2$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

Or

$$\begin{cases} x' = -y \\ y' = x \end{cases} \Rightarrow \begin{cases} x = y' \\ y = -x' \end{cases}$$

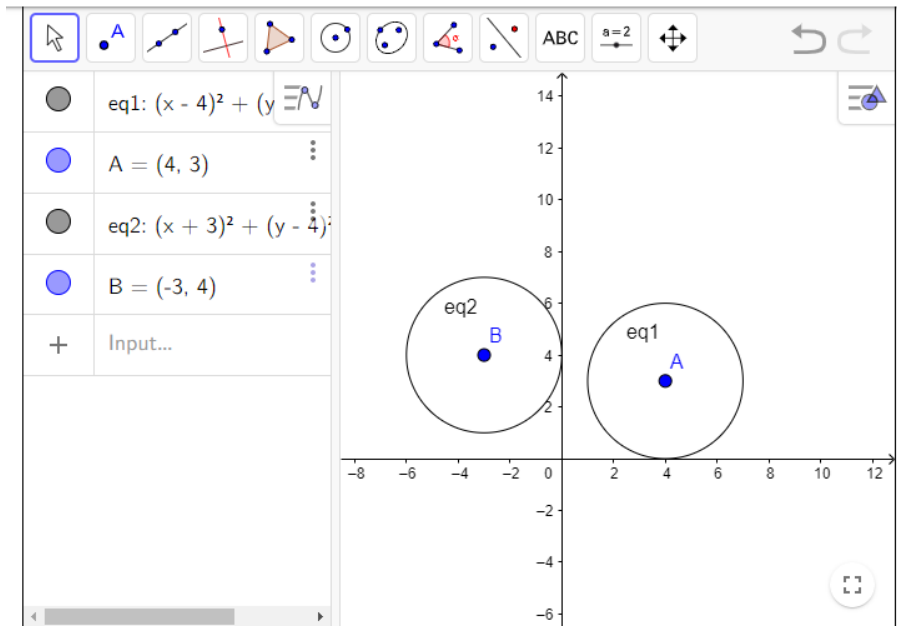
Putting these values of x and y in original equation of circle

$$(x - 4)^2 + (y - 3)^2 = 9$$

We get

$$(x' + 3)^2 + (y' - 4)^2 = 9$$

This is the equation of rotated circle with angle $\frac{\pi}{2}$ in anticlockwise direction.



Work to do:

Q4. Let $(x - 4)^2 + (y - 3)^2 = 9$ be a circle. Find the equation of circle after rotating it through an angle of $180^\circ, 270^\circ$ in clockwise direction about origin.

Example 4: Let $\frac{x^2}{(4)^2} + \frac{y^2}{(3)^2} = 1$ be an ellipse. Find the equation of ellipse after rotating it through an angle of 90° in anticlockwise direction about origin.

Solution: The transformation of rotation in anticlockwise direction is

$$T(\vec{x}) = A\vec{x} \quad : \forall \vec{x} \in R^2$$

$$T(\vec{x}) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \vec{x}$$

$$\text{As } \theta = 90^\circ, \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

$$\text{Hence,} \quad \begin{cases} x' = -y \\ y' = x \end{cases} \quad \text{or} \quad \begin{cases} x = y' \\ y = -x' \end{cases}$$

Put these values of x and y in original equation of ellipse, we get the rotated ellipse with angle $\frac{\pi}{2}$ in anticlockwise direction as

$$\frac{x'^2}{9} + \frac{y'^2}{16} = 1$$

For plotting we can neglect the dash (') from our rotated equation of ellipse.

Original Ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

Major axis is along x-axis

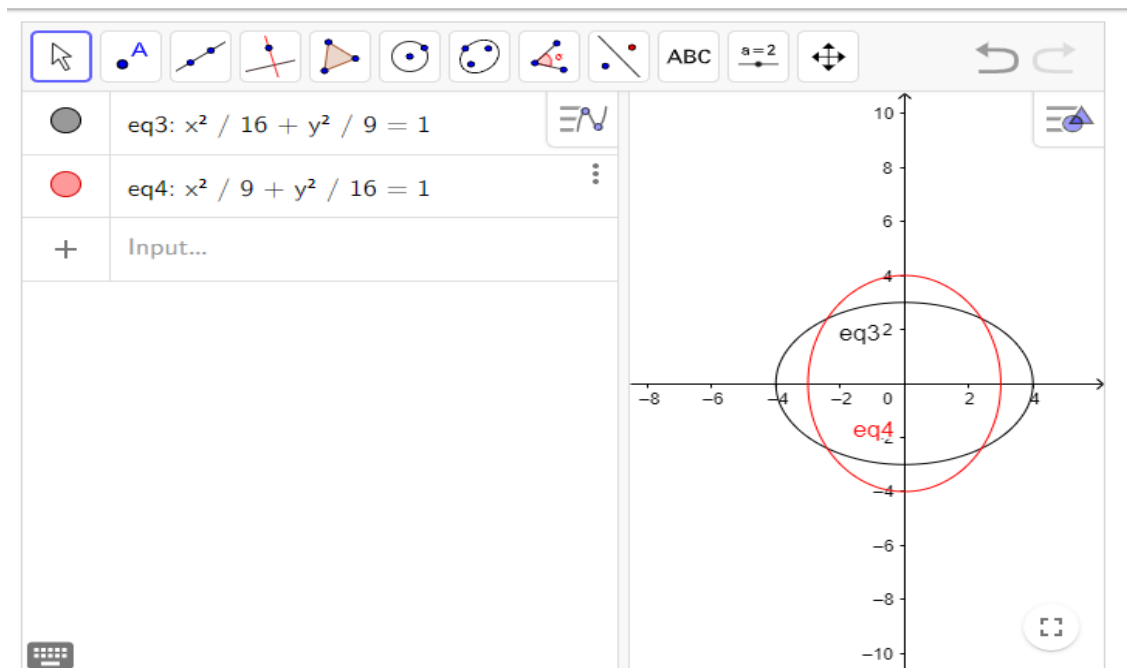
$$a = \pm 4, \quad b = \pm 3$$

Rotated Ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

Major axis is along- yaxis

$$a = \pm 3, \quad b = \pm 4$$



Q5. Let $\frac{x^2}{9} + \frac{y^2}{16} = 1$ be an ellipse. Find the equation of ellipse after rotating it through an angle of 180° in anticlockwise direction about origin.